

Activity 5: AI-Powered Project Discovery Challenge

Purpose

This capstone activity consolidates all the concepts covered during Day 1. Learners will use Generative AI to analyze the SmartBank Net Banking Application, generate software engineering artifacts, and practice effective prompt engineering techniques. The outputs produced in this activity will serve as reference material for future coding and development activities.

Pre-requisites

- Completion of Activities 1–4
 - ChatGPT or Oracle Chat AI
 - Internet connection
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Estimated Duration

45 Minutes

Scenario

Your project kickoff meeting is scheduled for tomorrow.

Before development begins, your Project Manager has asked every team to prepare a **Project Discovery Document** that provides a high-level understanding of the SmartBank application.

Instead of manually creating the document, your team will leverage Generative AI to prepare the first draft. However, as software engineers, you are responsible for reviewing, refining, and validating all AI-generated content before submission.

Tasks

Complete the following tasks in sequence.

Task 1: Generate a Project Overview

Prompt

Act as a Senior Solution Architect.

Create a Project Overview document for the SmartBank Net Banking Application.

Include:

- Project Vision
- Business Objectives
- Scope of the Application
- Expected Benefits
- Key Stakeholders

Present the information using headings and bullet points.

Task 2: Create a Functional Module Summary

Prompt

Based on the SmartBank application, prepare a Functional Module Summary.

For each module include:

- Module Name
- Purpose
- Major Features
- Dependencies

Present the output as a table.

Task 3: Generate User Personas

Prompt

Create user personas for the SmartBank application.

Include the following users:

- Retail Customer
- Bank Administrator
- Customer Support Executive

For each persona provide:

- Goals
 - Responsibilities
 - Features Frequently Used
 - Challenges
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Task 4: Generate Business Rules

Prompt

Generate business rules for the SmartBank application.

Include rules for:

- User Login
- Fund Transfer
- Beneficiary Management
- Bill Payments

Present the rules in a numbered list.

Task 5: Generate a Glossary

Prompt

Create a glossary for the SmartBank project.

Include common banking and software terms such as:

- Beneficiary
- IFSC
- Transaction
- REST API
- Authentication
- Dashboard
- Session
- Encryption

Provide a simple definition for each term suitable for a fresher software engineer.

Task 6: Generate a Project FAQ

Prompt

Create a Frequently Asked Questions (FAQ) document for new developers joining the SmartBank project.

Include at least ten questions and answers related to:

- Project Overview
- Development Process
- Banking Concepts

- Technologies
 - AI Usage During Development
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Task 7: Refine and validate the AI Output

Review all AI-generated content and identify:

- Any assumptions made by the AI.
- Any information that should be verified.
- Any sections that need improvement.
- Any sensitive information that should not be included in project documentation.

Make the necessary refinements before finalizing the document.

Expected Deliverables

By the end of this activity, you should have:

- Project Overview
 - Functional Module Summary
 - User Personas
 - Business Rules
 - Project Glossary
 - Developer FAQ
 - Reviewed and Refined AI Outputs
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Challenge Task

Your instructor will assign one of the SmartBank modules (e.g., Login, Fund Transfer, Beneficiary Management, Dashboard, or Bill Payments) to your team.

Using everything learned during Day 1, create a **master prompt** that generates a complete project analysis for the assigned module.

Your prompt should:

- Assign a role to the AI.
- Provide sufficient project context.
- Clearly define the objective.
- Specify the expected output format.
- Request structured, well-organized responses.
- Instruct the AI to avoid making unsupported assumptions.

- Request the AI to clearly identify any assumptions or information that requires verification.

Execute your prompt, review the generated output, and refine it until it meets professional standards.